

MA.912.AR.2.8

Benchmark

MA.912.AR.2.8 Given a mathematical or real-world context, graph the solution set to a two-variable linear inequality.

Benchmark Clarifications:

Clarification 1: Instruction includes the use of standard form, slope-intercept form and point-slope form and any inequality symbol can be represented.

Clarification 2: Instruction includes cases where one variable has a coefficient of zero.

Connecting Benchmarks/Horizontal Alignment

- MA.912.AR.4.1, MA.912.AR.4.3
- MA.912.AR.9.1, MA.912.AR.9.4, MA.912.AR.9.6

Terms from the K-12 Glossary

- Coordinate Plane
- Linear Expression

Vertical Alignment

Previous Benchmarks

- MA.8.AR.3.4

Next Benchmarks

- MA.912.AR.3.10
- MA.912.AR.9.8

Purpose and Instructional Strategies

In grade 8, students graphed linear two-variable equations. In Algebra I, students graph the solution set to a two-variable linear inequality. In later courses, students will solve problems involving linear programming and will graph the solutions sets of two-variable quadratic inequalities.

- Instruction includes the use of linear inequalities in standard form, slope-intercept form and point-slope form. Include examples in which one variable has a coefficient of zero such as $x < -\frac{17}{5}$.
- Instruction includes the connection to graphing solution sets of one-variable inequalities on a number line; recognizing whether the boundary line should be dotted (exclusive) or solid (inclusive). Additionally, have students use a test point to confirm which side of the line should be shaded (*MTR.6.1*).
- Students should recognize that the inequality symbol only directs where the line is shaded (above or below) for inequalities when in slope-intercept form. Students shading inequalities in other forms will need to use a test point to determine the correct half-plane to shade.

Common Misconceptions or Errors

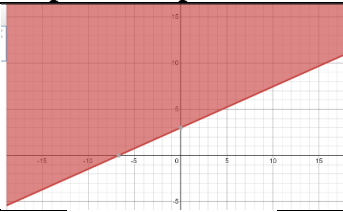
- Students often choose to shade to wrong half-plane when graphing two-variable linear inequalities.
- Students may think that the inequality symbol's orientation always determines the side of the line to shade.
 - For example, students may say that inequalities with a less than symbol should be shaded below the line while inequalities with a greater than symbol should be shaded above the line. This typically happens after graphing multiple inequalities in slope-intercept form. To address this, provides counterexamples to this such as $3x - 2y < 15$ or $-4x - 7 \geq y$. Use these counterexamples to emphasize the benefit of using a test point to confirm the direction of shading.

Strategies to Support Tiered Instruction

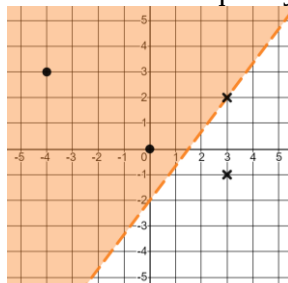
- Instruction includes opportunities to use a highlighter to identify the phrases “is less than,” “is greater than,” “is less than or equal to,” and “is greater than or equal to” when writing inequalities.
- Teacher provides instruction modeling how to correctly identify the solution set of a linear inequality given in slope-intercept form. After graphing, students can circle the y -intercept. If the inequality is in form $y < mx + b$ or $y \leq mx + b$, the solution set is the half-plane that contains the y -axis values below the y -intercept. If the inequality is in form $y > mx + b$ or $y \geq mx + b$, the solution set is the half-plane that contains the y -axis values above the y -intercept.
- Instruction includes opportunities to graph the boundary line of a system of inequalities, based on an inaccurate translation from word problem. To assist in determining the boundary line for the system, students can create a graphic organizer like the one below.

<	≤	>	≥
Less than	Less than and equal to	Greater than	Greater than and equal to
Fewer than	Maximum	More than	Minimum
Does not include	At most	Does not include	No less than
Dashed line	Solid Line	Dashed line	Solid Line

- Instruction includes making the connection between the algebraic and graphical representations of a two-variable linear inequality and its key features.
 - For example, teacher can provide a graphic organizer such as the one below.

Algebraic Representation	Graphical Representation
$y \geq 3 + \frac{4}{9}x$	
The y-intercept is located at the point (0,3).	
The slope of the boundary line is $\frac{4}{9}$.	From any point on the boundary line, the next point can be found by moving up/down 4 units and then moving right/left 9 units.
≥	The boundary line is solid with the with the solution set shaded above the boundary line.

- Instruction includes opportunities to identify a test point to plug into an inequality. It is usually easiest to use the origin (0,0) as it makes mental calculations easier. If the point selected creates a true statement, their inequality is true and they should shade in the half-plane containing that point. If it creates a false statement, they should shade in the half-plane not containing that point. By using a test point, students avoid the mistake of thinking that the direction of the inequality determines the shading.
 - For example, the points (−4,3), (0,0), (3,2) and (3,−1) were used to determine where to shade for the inequality $4x - 3y < 6$ shown below.



Instructional Tasks

Instructional Task 1 (MTR.7.1)

Penelope is planning to bake cakes and cookies to sell for an upcoming school fundraiser. Each cake requires $1\frac{3}{4}$ cups of flour and each batch of cookies requires $2\frac{1}{4}$ cups of flour.

Penelope bought 3 bags of flour. Each bag contains around 17 cups of flour.

Part A. Assuming she has all the other ingredients needed, create a graph to show all the possible combinations of cakes and batches of cookies Penelope could make.

Part B. Create constraints for this given situation.

Instructional Items

Instructional Item 1

Graph the solution set to the inequality $y + 3 > -2(x - 2)$.

**The strategies, tasks and items included in the B1G-M are examples and should not be considered comprehensive.*